
Phase 1: Requirements & Planning

1. Gather Requirements

- Define business goals and expected user experience.
- List must-have features (panels, spam filtering, scalability, security, etc.).
- Benchmark Zoho Mail and other futuristic platforms for inspiration.

2. Resource Planning

- Estimate hardware/cloud needs for 100,000 users.
 - Plan for future scalability (storage, compute, network).
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Phase 2: Architecture & Technology Selection

1. Choose Core Components

- SMTP: Postfix (recommended)
- IMAP/POP3: Dovecot.
- Spam Filtering: Rspamd (modern, fast).
- Database: MariaDB for user/account storage.
- Webmail: Roundcube.
- Admin Panels: Mailu
- Management panel:
- Storage: GlusterFS or Ceph for distributed storage.
- Load Balancer: Nginx.

2. Design System Architecture

- Draw diagrams for mail flow, storage, user management, and security.
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Phase 3: Infrastructure Setup

1. Provision Servers

- Install Ubuntu 24.04 on all nodes.
- Configure networking, DNS, firewalls.

2. Set Up Storage

- Deploy distributed file system (Ceph).
- Configure backup storage.

3. SSL/TLS Setup

- Obtain certificates (Let's Encrypt).
 - Configure SSL for all services.
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Phase 4: Core Email System Installation

1. Install Mail Server

- Install and configure Postfix (SMTP).
- Install and configure Dovecot (IMAP/POP3).
- Define virtual domains, users, and quotas.

2. Configure Domain Security

- Set up DKIM (sign outgoing mail), SPF, DMARC records.
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Phase 5: Spam Filtering & Security

1. Install and Integrate Spam Filter

- Install Rspamd.
- Integrate with Postfix and Dovecot.

2. Configure Policies

- Fine-tune spam rules, whitelist/blacklist.
- Set up quarantine and reporting.

3. Server Hardening

- Install fail2ban for brute-force protection.

- Regular OS and package updates.
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Phase 6: Web Panels & User Management

1. Deploy Customer Panel

- Install/configure webmail (Roundcube).
- Customize branding and UX.

2. Deploy Admin Panel

- Install Mailu or custom admin panel.
- Set up user/account management, monitoring, reporting.

3. Deploy Separate Management Panel

- For super-admins: advanced controls, billing, analytics.
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Phase 7: Scalability & Performance

1. Load Balancing

- Configure Nginx for web and mail traffic.

2. Optimize Database and Queues

- Tune MariaDB for high concurrency.
- Monitor and manage mail queues.

3. Horizontal Scaling

- Add more mail nodes as needed.
 - Partition users across servers/storage.
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Phase 8: Advanced Features & Integrations

1. Enable 2FA

- Integrate two-factor authentication for webmail and panels.

2. Contacts, Calendar, Tasks

- Integrate Nextcloud or similar.
 - 3. Advanced Search**
 - Set up Elasticsearch for mail search.
 - 4. API Access**
 - Provide endpoints for external integrations.
 - 5. Mailing Lists, Aliases, Auto-responders**
 - Configure in admin panel.
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Phase 9: Monitoring, Backup & Maintenance

- 1. Monitoring**
 - Install Prometheus, Grafana, or Zabbix for health monitoring.
 - Set up log management and alerting.
 - 2. Backup & Archiving**
 - Integrate Mailpiler or other archiving tools.
 - Schedule regular backups and test restores.
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Phase 10: Testing & Documentation

- 1. Functional and Security Testing**
 - spam filtering, Test mail deliverypanel, functionalities.
 - Simulate load for 100,000 users.
 - 2. Documentation**
 - Write installation/configuration guides.
 - Provide user/admin manuals.
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Phase 11: Go-live & Support

- 1. Deployment**

- Roll out to pilot users, then full launch.
- Monitor and optimize based on feedback.

2. Support

- Set up helpdesk or support portal.
- Plan for regular updates and improvements.